AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Swayam D. Gode
Roll No. :	70
Experiment No.:	2
Title of the Experiment:	Apply Descriptive Statistics and Data Visualization Techniques for Exploratory Data Analysis
Date of Performance:	28/07/2026
Date of Submission:	04/08/2026

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : Signature :

Date :

Aim: To apply descriptive statistical measures and data visualization techniques to summarize, analyze, and interpret the distribution and relationships present in the Pima Indians Diabetes Dataset.

Objectives

After completing this experiment, the student will be able to:

- Apply descriptive statistical and visualization techniques to explore the characteristics of a real-world healthcare dataset.
- Interpret distributions, variability, relationships, and patterns obtained from statistical analysis and visualizations.

Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. Matplotlib
6. Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for female patients and is used to study factors associated with diabetes.

The dataset contains 768 observations and 8 input attributes, along with a binary outcome variable indicating diabetes status.

Attribute	Description

Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Triceps skin fold thickness
Insulin	2-Hour serum insulin
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes pedigree function
Age	Age in years
Outcome	1 = Diabetic, 0 = Non-diabetic

Procedure

1. Load the Pima Indians Diabetes Dataset into a Pandas DataFrame.
2. Inspect the dataset structure, dimensions, column names, and data types.
3. Calculate descriptive statistics including mean, median, mode, minimum, maximum, variance, and standard deviation for selected numerical variables.
4. Analyze the distribution of variables using histograms and boxplots.
5. Analyze the frequency of the Outcome variable using a bar chart.
6. Generate scatter plots to study relationships such as Glucose vs. BMI and Age vs. Glucose.
7. Generate a pair plot for selected numerical variables.
8. Interpret the statistical summaries and visualizations and record important observations. Description of the Experiment

Descriptive statistics summarize the important characteristics of a dataset using numerical measures, while data visualization presents these characteristics graphically.

The experiment follows the workflow: Dataset, Descriptive Statistics, Distribution Analysis, Visualization, Relationship Analysis, Interpretation,

Students will use the Pima Diabetes Dataset to analyze the central tendency, variability, distributions, and relationships among health-related variables.

Detailed Description of the Statistical techniques:

Mean, Median, and Mode

These measures describe the central or typical value of a dataset.

$$\bar{x} = \frac{1}{n} \quad x$$

The median is the middle value, while the mode is the most frequently occurring value.

Range, Variance, and Standard Deviation These

measures describe data variability.

$$Range = Maximum - Minimum$$

$$s = \frac{\sum(x - \bar{x})}{n - 1}$$

$$s = \sqrt{s}$$

Histogram

Used to study the distribution, spread, and shape of numerical variables such as Glucose, BMI, and Age.

Boxplot

Used to visualize the median, quartiles, spread, and potential outliers.

$$IQR = Q - Q$$

Bar Chart

Used to display the frequency of categorical or binary variables such as Outcome.

Scatter Plot

Used to examine the relationship between two numerical variables, such as:

- Glucose and BMI
- Age and Glucose

Pair Plot

Used to simultaneously visualize pairwise relationships and distributions among multiple numerical variables.

CODE:-

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")

df = pd.read_csv("diabetes.csv")
```

```
print("=" * 60)
print("PIMA INDIANS DIABETES DATASET")
print("=" * 60)

print("\nFirst 5 Rows")
print(df.head())

print("\nLast 5 Rows")
print(df.tail())

print("\nDataset Shape")
print(df.shape)

print("\nColumn Names")
print(df.columns.tolist())

print("\nData Types")
print(df.dtypes)

print("\nDataset Information")
df.info()

print("\nMissing Values")
print(df.isnull().sum())
```

```
print("\n" + "=" * 60)
print("DESCRIPTIVE STATISTICS")
print("=" * 60)

print("\nSummary Statistics")
print(df.describe())

print("\nMean")
print(df.mean(numeric_only=True))

print("\nMedian")
print(df.median(numeric_only=True))

print("\nMode")
print(df.mode().iloc[0])

print("\nMinimum")
print(df.min(numeric_only=True))

print("\nMaximum")
print(df.max(numeric_only=True))

print("\nVariance")
print(df.var(numeric_only=True))

print("\nStandard Deviation")
```

```
print(df.std(numeric_only=True))

print("\nRange")
range_values = df.max(numeric_only=True) - df.min(numeric_only=True)
print(range_values)

df.hist(figsize=(15, 12), bins=20, edgecolor='black')
plt.suptitle("Histograms of Numerical Variables", fontsize=16)
plt.tight_layout()
plt.show()

plt.figure(figsize=(14, 7))
sns.boxplot(data=df)
plt.xticks(rotation=45)
plt.title("Boxplots of Numerical Variables")
plt.show()

columns = [
    "Pregnancies",
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
    "BMI",
    "DiabetesPedigreeFunction",
    "Age"
```

```
]

for col in columns:
    plt.figure(figsize=(5, 5))
    sns.boxplot(y=df[col], color="skyblue")
    plt.title(f"Boxplot of {col}")
    plt.show()

plt.figure(figsize=(6, 5))
sns.countplot(x="Outcome", data=df)
plt.title("Frequency of Outcome")
plt.xlabel("Outcome (0 = Non-Diabetic, 1 = Diabetic)")
plt.ylabel("Count")
plt.show()

plt.figure(figsize=(7, 5))
sns.scatterplot(
    x="Glucose",
    y="BMI",
    hue="Outcome",
    data=df
)

plt.title("Glucose vs BMI")
plt.show()
```

```
plt.figure(figsize=(7, 5))
sns.scatterplot(
    x="Age",
    y="Glucose",
    hue="Outcome",
    data=df
)

plt.title("Age vs Glucose")
plt.show()

pair_columns = [
    "Glucose",
    "BMI",
    "Age",
    "BloodPressure",
    "Outcome"
]

sns.pairplot(
    df[pair_columns],
    hue="Outcome",
    diag_kind="hist"
)

plt.show()
```

```
plt.figure(figsize=(10, 8))
sns.heatmap(
    df.corr(),
    annot=True,
    cmap="coolwarm",
    linewidths=0.5
)

plt.title("Correlation Matrix")
plt.show()

std_values = df.std(numeric_only=True)

highest_std_variable = std_values.idxmax()

print("\nVariable with Highest Standard Deviation")
print(highest_std_variable)
print("Standard Deviation =", std_values.max())

difference = abs(
    df.mean(numeric_only=True) -
    df.median(numeric_only=True)
)

print("\nDifference Between Mean and Median")
```

```
print(difference)

largest_difference = difference.idxmax()

print("\nVariable with Largest Mean-Median Difference")
print(largest_difference)

print("\nOutlier Detection using IQR")

for col in columns:

    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)

    IQR = Q3 - Q1

    lower_limit = Q1 - 1.5 * IQR
    upper_limit = Q3 + 1.5 * IQR

    outliers = df[
        (df[col] < lower_limit) |
        (df[col] > upper_limit)
    ]

    print(f"{col}: {len(outliers)} outliers")
```

```
print("\n" + "=" * 60)

print("OBSERVATIONS")

print("=" * 60)


print("1. Histograms show the distribution of numerical variables.")

print("2. Boxplots indicate outliers in Insulin, BMI, Pregnancies and DiabetesPedigreeFunction.")

print("3. Glucose has a positive relationship with diabetes outcome.")

print("4. Scatter plots show a weak positive relationship between Glucose and BMI.")

print("5. Correlation heatmap shows Glucose has one of the strongest correlations with Outcome.")


print("\nExperiment Completed Successfully.")
```

OUTPUT:-

```
============================================================
PIMA INDIANS DIABETES DATASET
============================================================

First 5 Rows
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72             35        0  33.6                     0.627   50        1
1            1       85             66             29        0  26.6                     0.351   31        0
2            8      183             64              0        0  23.3                     0.672   32        1
3            1       89             66             23       94  28.1                     0.167   21        0
4            0      137             40             35      168  43.1                     2.288   33        1

Last 5 Rows
     Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome
763           10      101             76             48      180  32.9                     0.171   63        0
764            2      122             70             27        0  36.8                     0.340   27        0
765            5      121             72             23      112  26.2                     0.245   30        0
766            1      126             60              0        0  30.1                     0.349   47        1
767            1       93             70             31        0  30.4                     0.315   23        0

Dataset Shape
(768, 9)
```

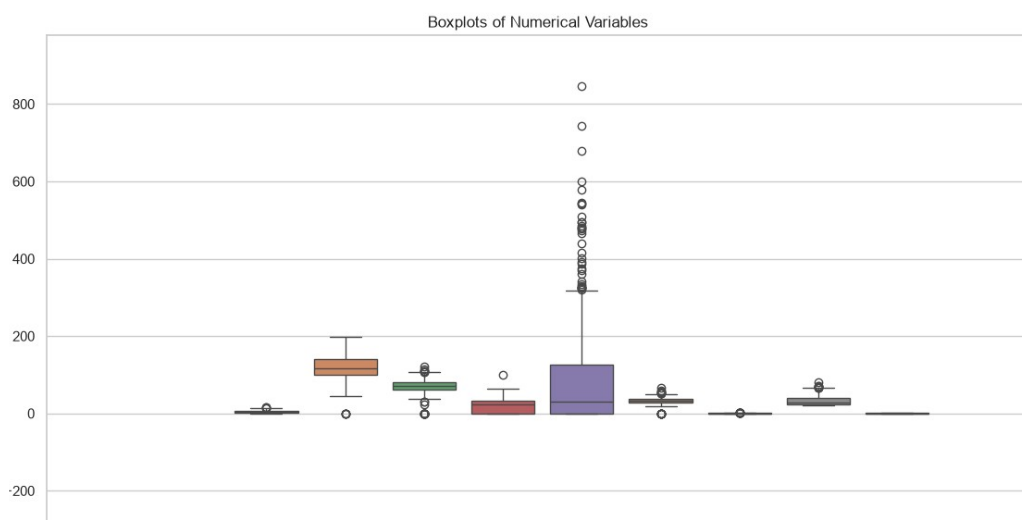
```
Column Names
['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome']

Data Types
Pregnancies                   int64
Glucose                       int64
BloodPressure                 int64
SkinThickness                 int64
Insulin                       int64
BMI                         float64
DiabetesPedigreeFunction    float64
Age                           int64
Outcome                       int64
dtype: object

Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```

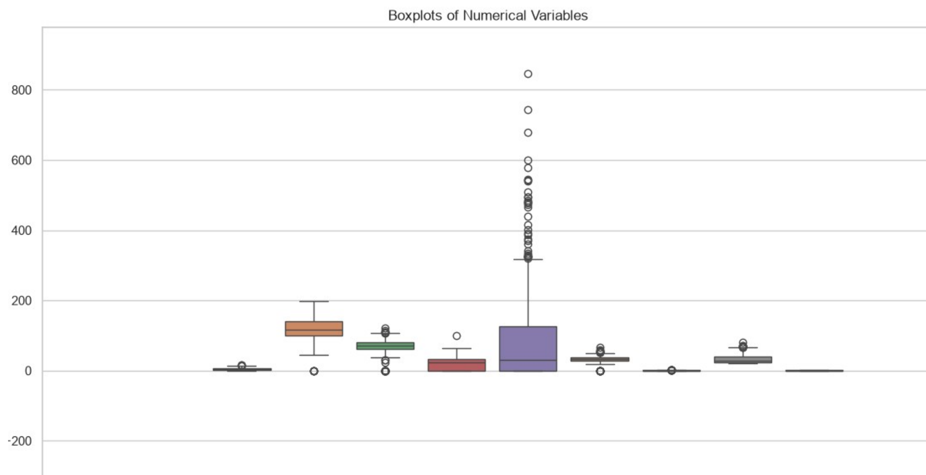
```
Missing Values
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

============================================================
DESCRIPTIVE STATISTICS
============================================================

Summary Statistics
```

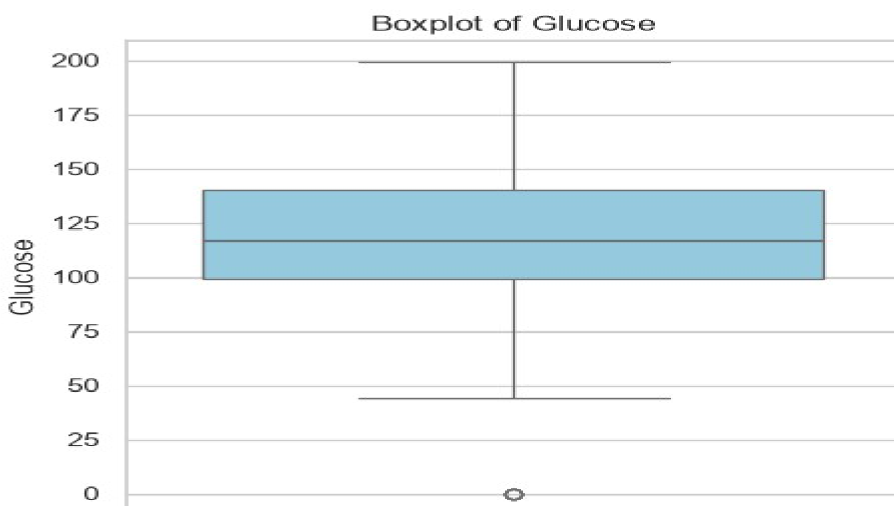
	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479	31.992578	0.471876	33.240885	0.348958
std	3.369578	31.972618	19.355807	15.952218	115.244002	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.078000	21.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000	27.300000	0.243750	24.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000	32.000000	0.372500	29.000000	0.000000
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	0.626250	41.000000	1.000000
max	17.000000	199.000000	122.000000	99.000000	846.000000	67.100000	2.420000	81.000000	1.000000

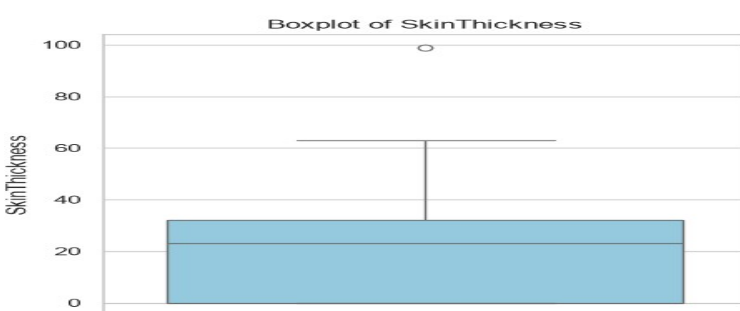
```
Mean
Pregnancies                    3.845052
Glucose                      120.894531
BloodPressure                 69.105469
SkinThickness                 20.536458
Insulin                       79.799479
BMI                           31.992578
DiabetesPedigreeFunction       0.471876
Age                           33.240885
Outcome                        0.348958
dtype: float64

Median
Pregnancies                    3.0000
Glucose                      117.0000
BloodPressure                 72.0000
SkinThickness                 23.0000
Insulin                       30.5000
BMI                           32.0000
DiabetesPedigreeFunction       0.3725
Age                           29.0000
Outcome                        0.0000
dtype: float64

Mode
Pregnancies                   1.000
Glucose                      99.000
BloodPressure                70.000
SkinThickness                 0.000
Insulin                       0.000
BMI                          32.000
DiabetesPedigreeFunction      0.254
Age                          22.000
Outcome                       0.000
Name: 0, dtype: float64
```

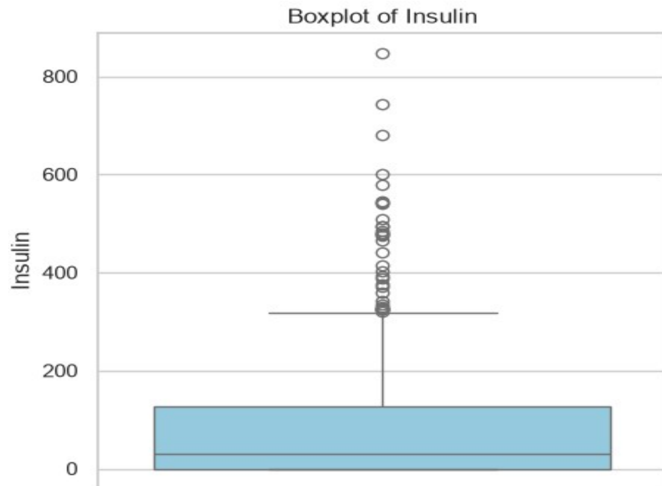
```
Minimum
Pregnancies                  0.000
Glucose                      0.000
BloodPressure                0.000
SkinThickness                0.000
Insulin                      0.000
BMI                          0.000
DiabetesPedigreeFunction     0.078
Age                         21.000
Outcome                      0.000
dtype: float64

Maximum
Pregnancies                  17.00
Glucose                     199.00
BloodPressure               122.00
SkinThickness                99.00
Insulin                     846.00
BMI                          67.10
DiabetesPedigreeFunction      2.42
Age                          81.00
Outcome                       1.00
dtype: float64

Variance
Pregnancies                    11.354056
Glucose                      1022.248314
BloodPressure                 374.647271
SkinThickness                 254.473245
Insulin                     13281.180078
BMI                            62.159984
DiabetesPedigreeFunction        0.109779
Age                           138.303046
Outcome                         0.227483
dtype: float64
```

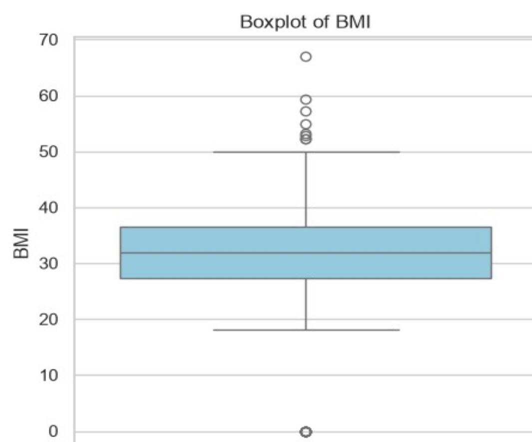


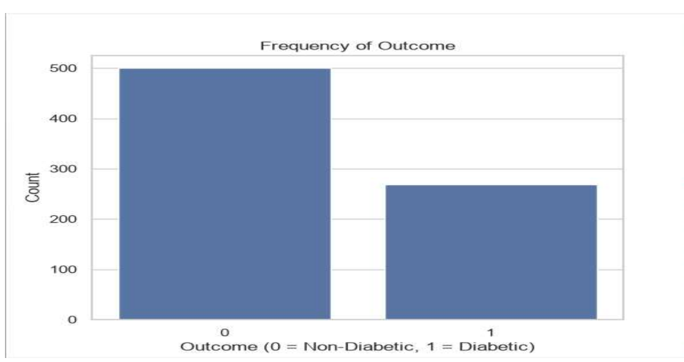

Boxplots of Numerical Variables
800
600
400
200
0
-200

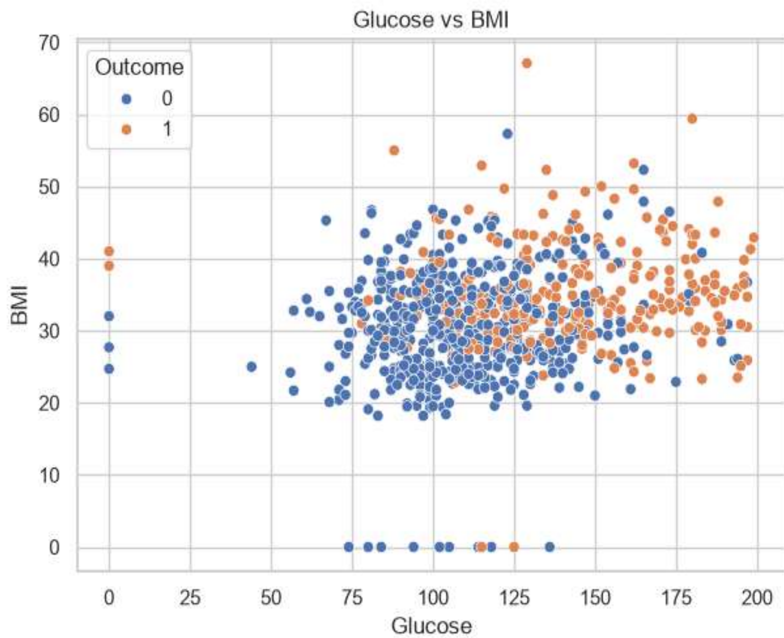

Boxplot of Glucose
200
175
150
125
100
75
50
25
0
Glucose

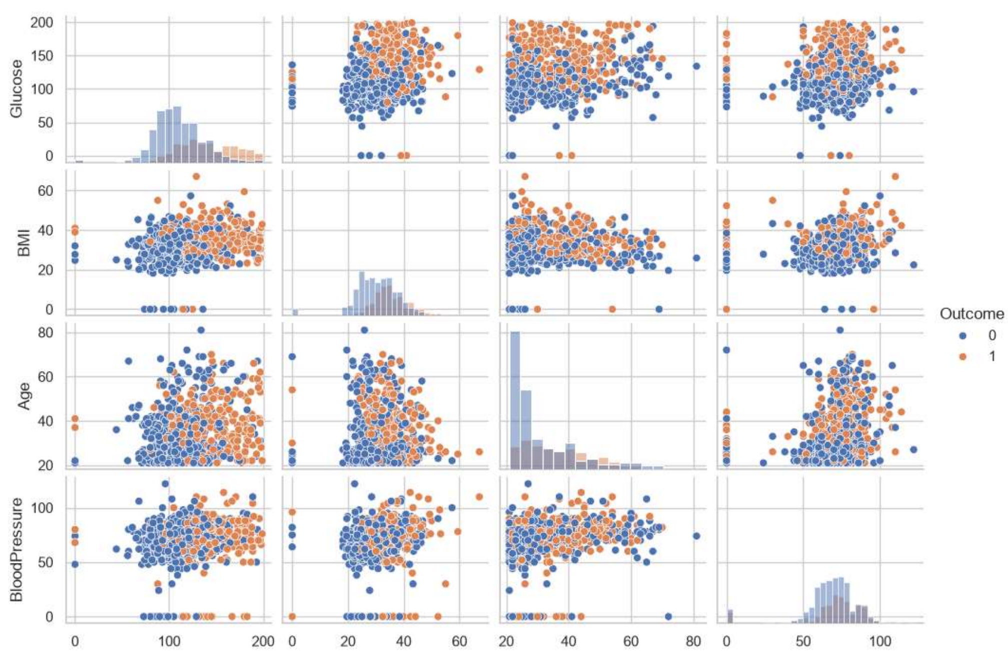

Boxplot of SkinThickness
100
80
60
40
20
0
SkinThickness

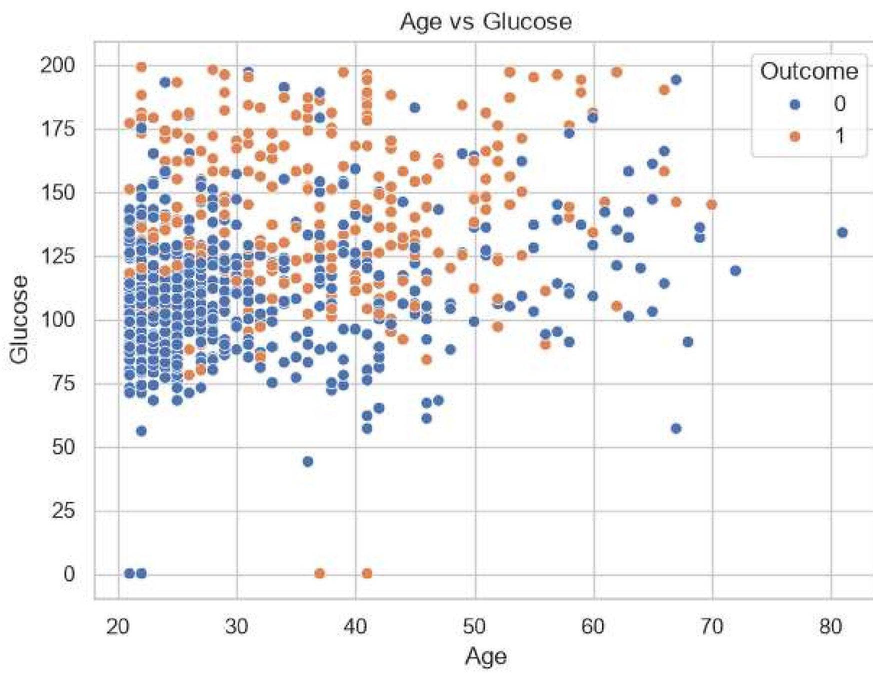

Boxplot of Insulin
Insulin
800
600
400
200
0

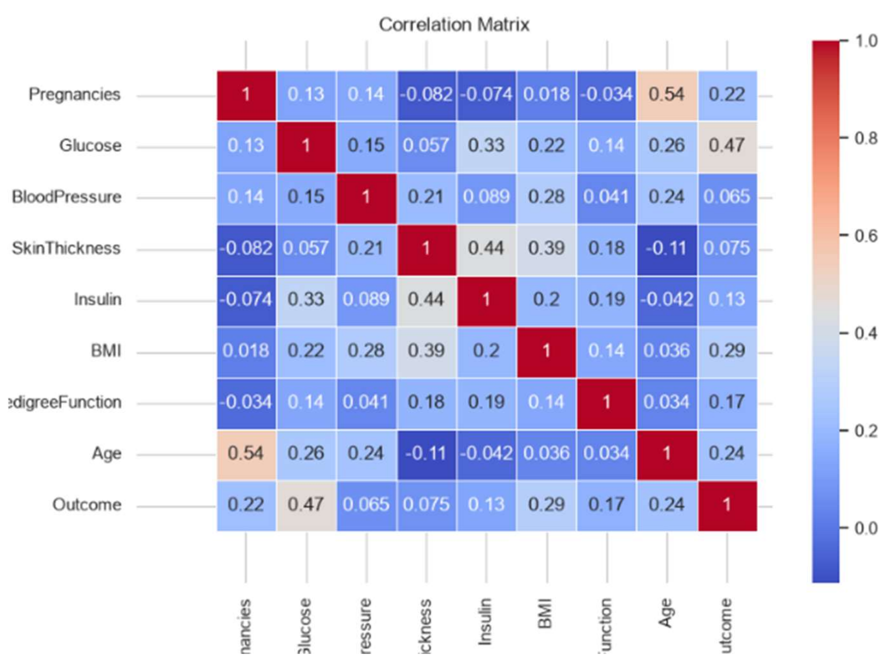

Boxplot of BMI
BMI
70
60
50
40
30
20
10
0


Frequency of Outcome
Count
500
400
300
200
100
0
0
1
Outcome (0 = Non-Diabetic, 1 = Diabetic)


Glucose vs BMI
Outcome
0
1
BMI
Glucose


Glucose
BMI
Age
BloodPressure
Outcome
0
1


Age vs Glucose
Glucose
Age
Outcome
0
1
200
175
150
125
100
75
50
25
0
20
30
40
50
60
70
80


Correlation Matrix
Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
edigreeFunction
Age
Outcome
1.0
0.8
0.6
0.4
0.2
0.0

Conclusion

Descriptive statistical measures and visualization techniques were applied to the Pima Indians Diabetes Dataset to understand the distribution, variability, and relationships among the variables.

Questions to be Answered by Students

Based on the standard **Pima Indians Diabetes Dataset (768 records)** and the exploratory analysis performed, the answers are:

1. Which numerical variable has the highest standard deviation?

The Insulin variable has the highest standard deviation because its values are spread over a very wide range, with several extremely high values (outliers).

2. Which variable shows the greatest difference between its mean and median?

The Insulin variable shows the largest difference between its mean and median. This occurs because the distribution is highly right-skewed, with many zero values (representing missing/invalid measurements) and a few very large insulin values that increase the mean.

3. What relationship can be observed between Glucose and BMI from the scatter plot?

The scatter plot indicates a weak to moderate positive relationship between Glucose and BMI. Patients with higher BMI tend to have higher glucose levels, although the relationship is not perfectly linear and there is considerable variability.

4. State two important insights obtained from the analysis.

1. Glucose is the strongest indicator of diabetes, with diabetic patients generally having significantly higher glucose levels than non-diabetic patients.
2. Several medical variables (Glucose, BloodPressure, SkinThickness, Insulin, and BMI) contain zero values, which likely represent missing or invalid measurements and should be handled before building predictive models.

5. Which vairable(s), if any, contain potential outliers? How were these identified?

Potential outliers were found in the following variables:

- Insulin
- BMI

- DiabetesPedigreeFunction
- Pregnancies
- (Some analyses also identify a few outliers in SkinThickness and Age.)

These outliers were identified using:

- Boxplots, which visually highlight observations beyond the whiskers.
- The Interquartile Range (IQR) method, where values less than Q1 − 1.5 × IQR or greater than Q3 + 1.5 × IQR are considered potential outliers.

6. State two important insights obtained from your exploratory statistical analysis of the dataset.

1. The dataset contains 768 observations, with 8 numerical predictor variables and 1 binary outcome variable, making it suitable for diabetes prediction and classification tasks.
2. The exploratory analysis revealed invalid zero values and several outliers, indicating that data preprocessing (handling missing values and evaluating outliers) is an essential step before applying statistical or machine learning models.